LLMs: Working, Limitations, and E-commerce Applications

Task 1: Concept Explanation

1. How LLMs Process Input Text

• **Tokens:** LLMs first split input text into smaller units called tokens. These tokens are the basic units processed by the model.

• **Embeddings:** Each token is converted into a numerical vector called an embedding. Embeddings capture semantic meaning and relationships between words.

2. How Tokens and Embeddings Help Generate Responses

The model uses token embeddings to identify patterns and relationships in the input. By analyzing these representations with attention mechanisms, it predicts the most likely next token repeatedly until a complete response is generated.

Task 2: Critical Thinking

1. Explain why LLMs do NOT truly "understand" language like humans

LLMs do not truly understand language like humans. They learn statistical patterns from massive datasets rather than possessing consciousness, reasoning, emotions, or real-world understanding. They predict likely text instead of genuinely comprehending meaning.

2. Identify ONE situation where relying fully on LLM output can be risky

Risky situation: Using an LLM alone to approve or reject customer loans or detect financial fraud. Incorrect or hallucinated outputs could lead to unfair or costly decisions, so human review is essential.

Task 3: Application Decision

1. Suggest TWO tasks in e-commerce where LLMs are reliable

Reliable e-commerce tasks:

• Generating product descriptions and marketing content.

• Providing customer support by answering common questions, tracking orders, and explaining return policies.

2. Suggest ONE task where LLMs should NOT be used without human verification

Task requiring human verification:

• Making final decisions on refunds, high-value disputes, credit approval, or legal/compliance issues.